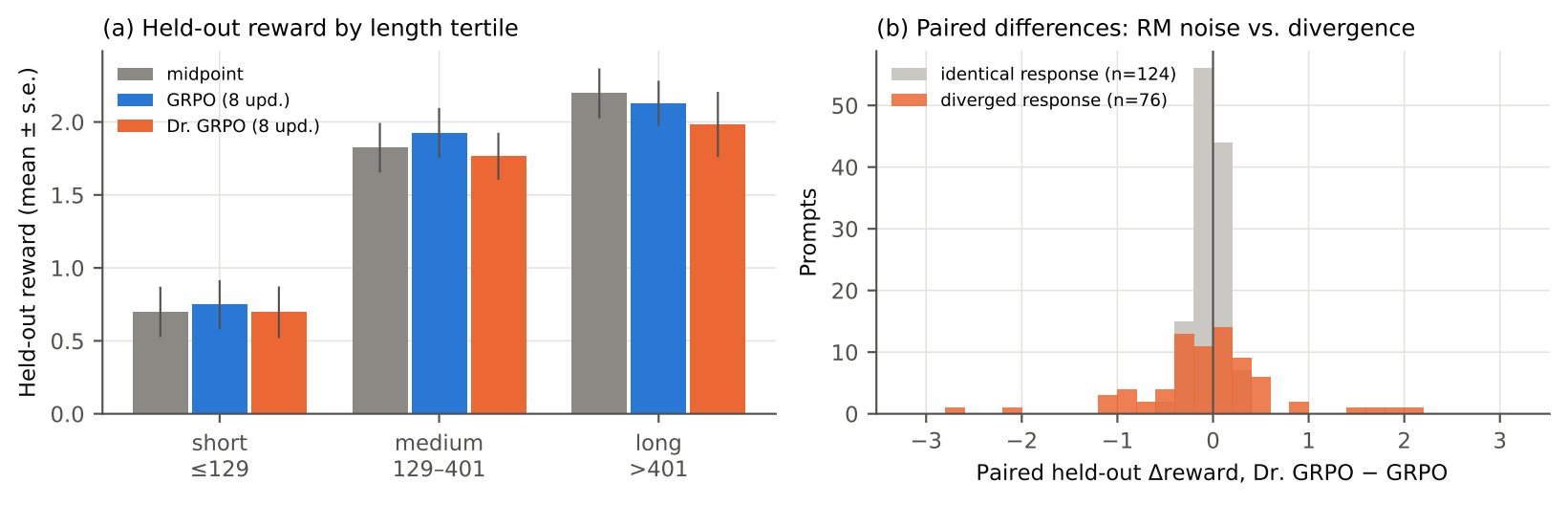
(a) Held-out reward by length tertile
midpoint
GRPO (8 upd.)
Dr. GRPO (8 upd.)
Held-out reward (mean ± s.e.)
0.0
0.5
1.0
1.5
2.0
short
≤129
medium
129–401
long
>401
(b) Paired differences: RM noise vs. divergence
identical response (n=124)
diverged response (n=76)
Prompts
0
10
20
30
40
50
−3
−2
−1
0
1
2
3
Paired held-out Δreward, Dr. GRPO − GRPO